Since *Bioecology* first appeared, there have been countless attempts to define what a biome is, and many different ways of classifying them. Biomes provide a simple way of understanding global vegetation patterns, but when looked at closely they present a crude way of grouping ecosystems. There is no single accepted classification system, and the only division everyone seems to agree on is that between terrestrial (land-based) and aquatic (water-based) biomes. Many of the same biomes crop up in most systems, such as the polar biome, tundra, rainforest, grasslands, and deserts, but there is no agreed definition and there are marked variations.



The Mongolian steppe belongs to the same grassland biome as the prairies in North America. Despite being on separate continents, they are linked by their climate, animals, and plants.